

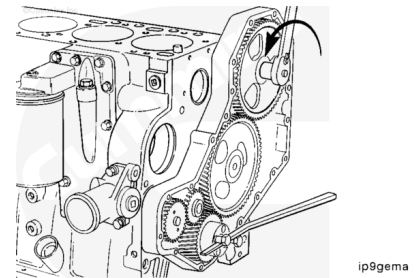
Trainings ▾

Remove

Front Gear Train

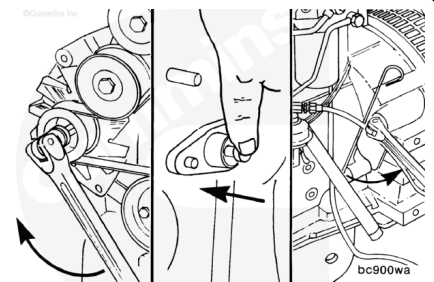
Remove the nut and washer from the fuel injection pump shaft.

Note : If necessary, insert two capscrews 180 degrees apart in the damper mounting locations in the nose of the crankshaft. Use an appropriate pry bar between the two capscrews to prevent the engine from turning when loosening the fuel pump gear mounting nut.



Note : Be sure to disengage the pin after locating top dead center.

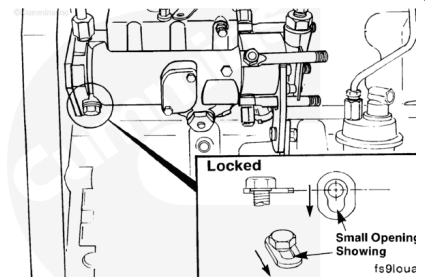
Locate top dead center for cylinder Number 1 by barring the engine slowly, while pushing in on the top dead center pin.



Lucas CAV DPA Pump and Delphi DP210

Loosen the CAV fuel injection pump lock screw and position the special washer; then tighten the lock screw against the pump drive shaft.

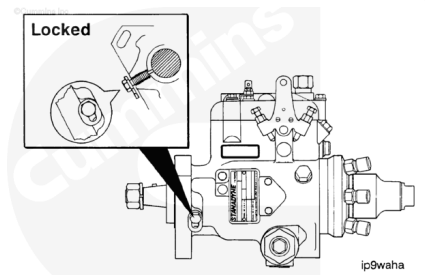
Torque Value: 7 n•m [62 in-lb]



Stanadyne DB4 Pump

Loosen the Stanadyne DB4 fuel injection pump lock screw and position the special washer. Tighten the lock screw until contact is made with the fuel injection pump drive shaft.

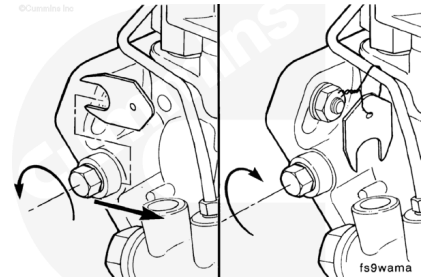
Torque Value: 12 n•m [106 in-lb]



The special washer on the Bosch® VE injection pump **must** be removed so the lock screw can be tightened against the drive shaft.

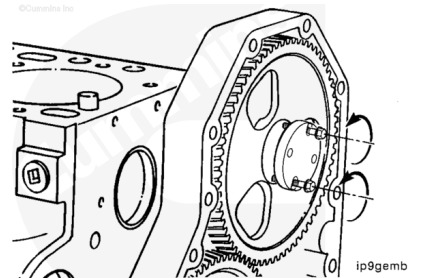
Torque Value: 30 n•m [22 ft-lb]

Note : Wire the washer to the pump.



Pull the fuel injection pump drive gear loose from the shaft.
Remove the fuel injection pump drive gear.

Use fuel pump gear puller, Part Number 3163381 or Part Number 3824469 with M8-1.25 x 50 capscrews, Grade 8.8 or equivalent.

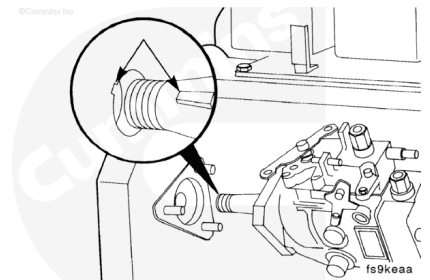


⚠ CAUTION ⚠

Do not drop drive gear key when removing the pump. Failure to do so can result in severe engine damage.

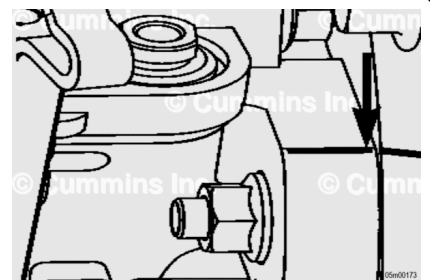
Remove the three mounting nuts and take off the fuel injection pump.

Note : Fuel pumps on engines designed to meet Tier 2/Stage II Industrial emissions levels have straight holes (not kidney slots) and do not use a timing key.

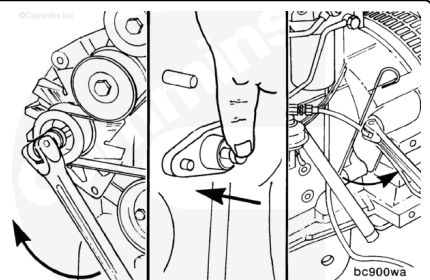


Rear Gear Train

Permanently mark the injection pump flange to match the mark on the fuel pump mounting plate.

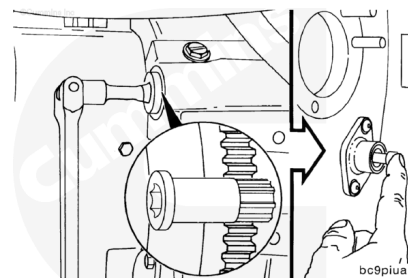


Unlike the front gear train engine, do **not** remove the timing pin until completion of the repair. Failure to do so can result in difficult reassembly and incorrect timing of the fuel pump.



Locate top dead center (TDC) for cylinder number 1 by barring the crankshaft slowly while pressing on the engine timing pin. Barring the engine is recommended from the flywheel on the rear of the engine.

Using the barring tool Part Number 3824591, rotate the crankshaft slowly while pressing on the engine timing pin to locate TDC for cylinder number 1.



⚠ CAUTION ⚠

Failure to properly torque the lock screw will result in improper timing of the pump during reassembly.

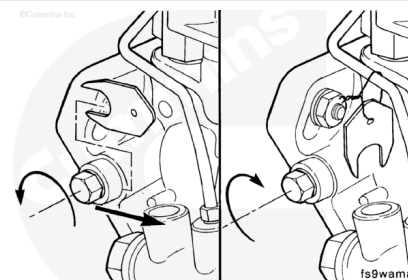
Bosch® VE

The special washer on the Bosch® VE injection pump **must** be removed so the lock screw can be tightened against the drive shaft.

Torque Value: 40 n•m [22 ft-lb]

Note : Wire the washer to the pump.

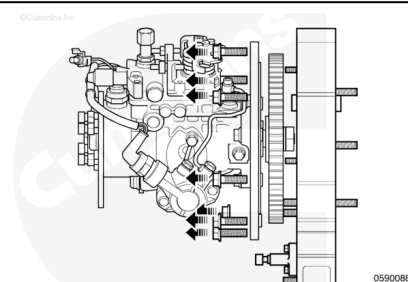
Note : The torque specification for the rear gear train engine is higher than the front gear train engine, to prevent rotation of the pump shaft during installation of the fuel pump gear retaining nut.



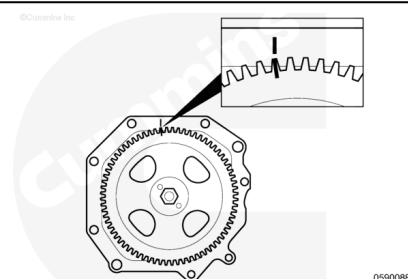
Remove the mounting nuts and bolts affixing the fuel pump mounting plate to the rear gear housing.

Take off the fuel injection pump, fuel pump mounting plate, and fuel pump gear as an assembly.

Note : Make sure the gear does not rotate during removal; failure to do so can result in incorrect timing of the pump during installation.



Mark a tooth on the fuel gear pump relative to the fuel pump mounting plate.



Last Modified: 27-Sep-2022
